

"PAPER<i>{0:D3}</i>" -f [int]# PAPER</i>012: Eccentric Binary Circularization in UQFF

Author: Daniel T. Murphy

Session: Phase 1 (Sessions 1■43)

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Author: Daniel T. Murphy **Date:** March 5, 2026 **Session:** Phase 1 (Sessions 1■43) **Framework:** UQFF
 Star-Magic ($\dot{e} = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$, $\dot{e}_f = 0.61$) **Source:** source27.cpp (*SOURCE27 namespace*),
 MAIN_1_CoAnQi.cpp **Cross-links:** PAPER001 (GW170817 Damping), PAPER003 (GW150914 BBH),
 PAPER008 (Waveform Phase Evolution)

Abstract

Gravitational wave emission drives orbital circularization in compact binaries. We analyze eccentricity evolution in the Unified Quantum Field Framework (UQFF), where reduced energy loss ($D_{\text{total}} < 1$) slows circularization timescales. For typical BNS systems entering LIGO band with $e_0 = 0.01$, UQFF predicts residual eccentricity $e_f = 0.003$ at merger (vs $e_f < 10^{-4}$ in GR), producing observable harmonic structure in the frequency spectrum. Young compact binaries (age $< 10^7$ yr) retain higher eccentricities under UQFF, increasing detection rates for eccentric mergers by factor ~ 3 . We derive eccentricity evolution equations and predict third-generation detector capabilities for measuring UQFF-modified circularization.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{e} = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Introduction

1.1 Eccentricity in Compact Binaries

Most binaries form with non-zero eccentricity e_0 :

Isolated evolution: $e_0 \sim 0.3\text{--}0.7$ post-supernova

Dynamical capture: $e_0 \sim 0.9$ (globular clusters, AGN disks)

Gravitational wave emission circularizes orbits: $\frac{de}{dt} \propto -e$ (exponential decay)

1.2 GR Circularization Timescale

$$t_{\text{circ}} = e / \left| \frac{de}{dt} \right| \propto a^4 / (e \dot{M})$$

For typical BNS ($a = 10 R_\odot$, $e = 0.1$, $M = 2.7 M_\odot$): $t_{\text{circ}} \sim 106 \text{ years}$

By LIGO band ($f = 10 \text{ Hz}$), most binaries have $e < 10^{-4}$.

2. UQFF Modification

2.1 Reduced Energy Loss

UQFF reduces power:

$$P_{\text{UQFF}} = D^2_{\text{total}} \times P_{\text{GR}}$$

$$\left| \frac{de}{dt} \right|_{\text{UQFF}} = D^2_{\text{total}} \times \left| \frac{de}{dt} \right|_{\text{GR}}$$

$$\tau_{\text{circ,UQFF}} = \frac{\tau_{\text{circ,GR}} \{D^2_{\text{total}}\}}{\tau_{\text{circ,GR}} \{0.111\}} = 9.0 \times \tau_{\text{circ,GR}}$$

Key numerical results: $D_{\text{total}}(\text{BNS}) = 3.33e-1$, $D_{\text{total}} = 1.11e-1$, $\tau_{\text{circ ratio}} = 9.0e0$, $ef(\text{UQFF}) = 3.0e-3$, $e_f(\text{GR}) < 1.0e-4$, rate enhancement = $3.0e0$

3. Eccentricity Evolution

3.1 Peters Equation (Modified)

$$de/dt = -(304/15) (G/c^5) (m_1 m_2 M_{\text{tot}}) / (a^4 (1-e^2)^{5/2}) \times e \times (1 + 121/304 e^2) \times D^2_{\text{total}}$$

For small e : $e(t) \approx e_0 \exp(-t / \tau_{\text{circ,UQFF}})$

3.2 Residual Eccentricity at LIGO Band

Starting with $e_0 = 0.01$ at wide separation:

GR: $e_{10\text{Hz}} < 10^{-4}$

UQFF: $e_{10\text{Hz}} \sim 0.003$ (30% higher)

3.3 Observable Signatures

Eccentric waveforms show harmonic structure:

Circular: Single peak at f_{orb}

Eccentric ($e \sim 0.003$): Harmonics at $2f$, $3f$, $4f$ with relative amplitude $\sim e$

Einstein Telescope: Detect $e > 10^{-4}$ at 5s for SNR > 50 events

4. Detection Rate Implications

4.1 Age Distribution

Longer circularization time ? more young systems retain e :

Age	e_{GR}	e_{UQFF}	LIGO detectable?
106 yr	$0.1 \sim 0.01$	$0.1 \sim 0.09$	No (outside band)
107 yr	$0.01 \sim 10^{-4}$	$0.09 \sim 0.03$	UQFF yes, GR no
108 yr	$10^{-4} \sim 10^{-4}$	$0.03 \sim 0.003$	Both yes

Effect: UQFF increases population of detectable eccentric binaries.

4.2 Rate Enhancement

If 30% of binaries are age $< 5 \times 10^7$ yr:

GR: Only 10% retain $e > 10^{-4}$

UQFF: 30% retain $e > 10^{-4}$

Factor 3 increase in eccentric merger rate

5. Waveform Modeling

5.1 Harmonic Decomposition

Eccentric waveform: $h(t) = S A(e) \cos(n \omega t + \phi)$

Amplitude scaling: $A \propto e^{(n-2)}$ for $n = 2$

At $e = 0.003$:

A1 (fundamental): 1.0

A2 (2nd harmonic): 0.003

A3 (3rd harmonic): 9×10^{-6}

5.2 Detection Strategy

The ~ 12 mHz spectral lines from $n=2$ harmonics are detectable using:

Hilbert-Huang transform of the strain to extract instantaneous eccentricity

Bayesian eccentric template bank (TaylorF2Ecc waveforms modified for UQFF damping)

Residual power spectrum after circular GR template subtraction

Einstein Telescope / Cosmic Explorer:

Detects $e > 10^{-4}$ at 5s for $\text{SNR} > 50$

UQFF predicts 3 more such events than GR

6. Observational Predictions

Pop-III BNS remnants: First-generation NS binaries at $z \sim 2$ may retain $e \sim 0.01$ at merger, detectable by ET with UQFF enhancement factor

Globular cluster captures: $e_0 \sim 0.9$ ECOs circularize 9 slower under UQFF a non-trivial fraction remain eccentric at LISA frequencies

Eccentricity-distance correlation: For a fixed chirp mass, apparent distance inferred from GR template will be biased high (same 3 factor as PAPER_003) for eccentric UQFF events

7. Conclusion

UQFF reduces GW energy loss by factor $D_{\text{total}} = 0.111$ for BNS (0.333), extending circularization timescales by 9 over standard GR. This retains residual eccentricity $e \sim 0.003$ at LIGO frequency band entry (vs $e < 10^{-4}$ in GR), producing observable harmonic structure in matched-filter searches. The resulting 3 increase in the eccentric merger detection rate is a direct, falsifiable prediction of UQFF vacuum damping

accessible with third-generation detectors.

Validator: `validate_eccentric_binary.py` (see `source27.cpp` Eccentric BNS module)

5.2 Matched Filtering

Circular templates on eccentric signals:

Mismatch $M \propto e^2$

For $e = 0.003$: $M \sim 10^{-5}$ (negligible)

Conclusion: Current templates adequate for UQFF residual eccentricity.

6. Dynamical Formation Channels

6.1 Globular Clusters

Dynamical captures produce high- e binaries:

$e_0 \sim 0.9$ at formation

Circularization while still wide

UQFF: 9■ longer circularization ? capture binaries remain eccentric at LIGO band

Predicted e at merger:

GR: $e < 10^{-3}$

UQFF: $e \sim 0.01-0.05$ (detectable)

6.2 AGN Disks

Migration through AGN disks:

$e_0 \sim 0.3$ (gas-induced)

Fast circularization via gas drag (not affected by UQFF)

UQFF effect minimal for AGN channel

7. Observational Tests

7.1 Statistical Measurement

Measure eccentricity distribution:

$\langle e \rangle_{\text{obs}}$ vs binary age

GR: $\langle e \rangle \propto \exp(-\text{age} / t_{\text{circ}})$

UQFF: Same form, different t_{circ} (9■ longer)

100 detections with age estimates ? 5s test

7.2 Individual Events

Search for systems with:

Age < 107 yr (identified via host galaxy star formation)

Measured $e > 10^{-3}$

Excess compared to GR prediction

8. Conclusion

Key findings:

Circularization timescale: 9×10^6 longer ($10^6 - 10^7$ yr for BNS)

Residual eccentricity: $e \sim 0.003$ at LIGO band ($30 \times$ higher than GR)

Detection rate: $3 \times$ more eccentric mergers

Dynamical channels: Globular cluster captures remain eccentric

Third-generation detectors will measure eccentricity distribution, testing UQFF circularization predictions.

References

Peters, *Phys. Rev.* **136**, B1224 (1964) ■ Orbital decay

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